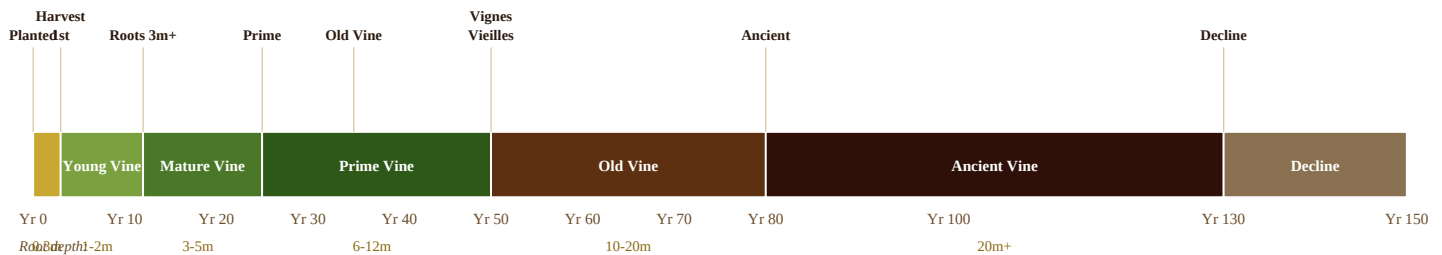


The Life of a Vine

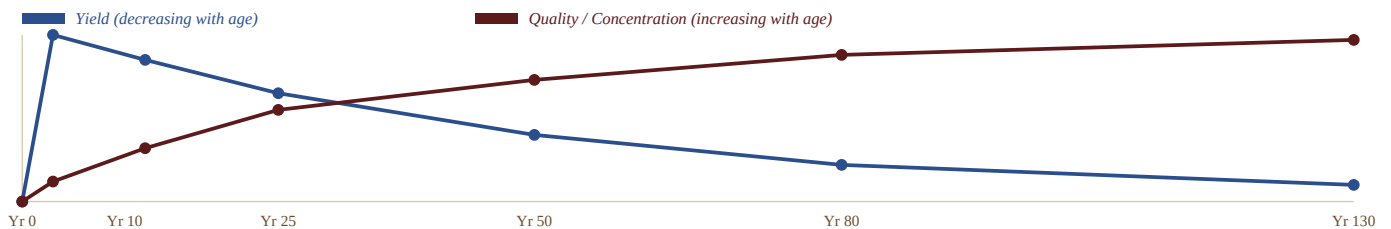
From cutting to ancient vine — what happens above and below ground at every stage · kazmicellars.com

A grapevine can live for 150 years or more. Its entire productive life is a journey of deepening roots, declining yields, and rising quality. The older the vine, the less fruit it makes — and the more extraordinary that fruit becomes. Understanding the vine's lifecycle is the foundation of understanding why old vine wines command the prices they do, and why great wineries treat their oldest blocks as irreplaceable living treasures.



* VV = Vieilles Vignes — no universal legal definition, generally accepted as 35–50+ years

Yield vs. Quality Over a Vine's Lifetime:



ESTABLISHMENT

Years 0–3

- Year 0 — A cutting or grafted rootstock is planted. Any flowers are removed — all energy goes to root establishment. The vine looks fragile; underground it is working hard.
- Years 1–2 — Root system expands slowly to 30–60cm depth. The vine is trained onto wires to form its permanent trunk structure. No commercial fruit is allowed.
- Year 3 — Roots reach 0.3–0.6m. First possible commercial yield, but quality is simple. Great estates wait until Year 4–5 before using the fruit in any bottle.

Root depth: 0.3–0.6m · Yield: Minimal · Quality: Too young · Key risk: Frost, drought

■ The patience invested in establishment determines the vine's entire productive life. Cutting corners here means a shorter-lived, less complex vine.

YOUNG VINE

Years 3–12

- Years 3–7 — The vine finds its rhythm. Roots push to 1–2m as the vine learns to access subsoil moisture. Yields are high — the winemaker must manage through green harvesting (removing bunches mid-season) to concentrate quality.
- Years 7–12 — Quality noticeably improves as roots deepen. The vine produces clean, bright, fruit-forward wine. Most entry-level estate wines come from this age range — accessible, approachable, and genuinely delicious, just not complex.

Root depth: 1–2m · Yield: 6–10+ tons/acre · Quality: Good, fruit-forward · Character: Bright and accessible

■ *Young vine wines are not lesser — they can be genuinely delicious. What they lack is the mineral depth that only comes from roots accessing ancient subsoil geology.*

MATURE VINE

Years 12–25

■ **Years 12–18** — The vine is fully established. Roots now reach 3–5m, accessing different soil layers and mineral profiles. Yields begin to self-regulate naturally — the vine produces less fruit without intervention.

■ **Years 18–25** — This is the workhorse of most premium estates. Fruit is consistent, quality is reliable, and the wine shows genuine complexity and a sense of place. The vast majority of the world's acclaimed wines come from vines in this age range.

Root depth: 3–5m · Yield: 4–7 tons/acre · Quality: Premium · Character: Complex, consistent, showing terroir

■ *If a winery labels a wine 'Estate' without further qualification, it most likely comes from mature vines in this window. This is the commercial sweet spot.*

PRIME VINE

Years 25–50

■ **Years 25–35** — A perceptible quality threshold is crossed. Many winemakers notice a clear jump in complexity at around 25–30 years. Roots reach 6–12m depth, accessing ancient subsoil minerals unavailable to younger vines.

■ **Years 35–50** — Yield reduction accelerates naturally. The vine invests more energy into fewer berries, creating greater concentration. Many estates segregate these blocks for their premium single-vineyard wines. The 'old vine' mineral depth and savory complexity begins to clearly distinguish these wines from younger blocks.

Root depth: 6–12m · Yield: 2–5 tons/acre · Quality: Very high to exceptional · Character: Mineral depth, concentration, clear sense of place

■ *The 35-year mark is where many estates apply 'old vine' or 'vieilles vignes' designations, though no country has a universal legal definition. Some use 25 years; others insist on 50+.*

OLD VINE — Vieilles Vignes

Years 50–80

■ **Years 50–65** — The vine achieves genuine self-sufficiency. Its deep root system (10–20m+) sources water from ancient aquifers and minerals from geological formations millions of years old. Yields are very low — often 2–3 tons/acre without any winemaker intervention.

■ **Years 65–80** — Old vines are gnarled and visually dramatic. The ratio of old wood to new growth is high, meaning the vine's stored energy is highly concentrated into very few berries. The wines have an intensity and complexity simply not achievable from younger vines — a mineral precision, structural elegance, and depth that requires time and deep roots to develop.

Root depth: 10–20m · Yield: 1.5–3 tons/acre · Quality: Exceptional · Character: Mineral precision, structural complexity, profound depth

■ *Famous old vine regions: Barossa Valley Grenache (80–100 yr), Priorat Carignan (70–90 yr), Champagne Pinot Meunier (70+ yr), old-vine Zinfandel in Lodi and Dry Creek Valley.*

ANCIENT VINE

Years 80–150+

■ **Years 80–120** — Ancient vines produce some of the world's greatest wines. Yields are extraordinarily low — sometimes under 1 ton/acre. The vine is nearly impossible to kill: its root system is so deep that drought and disease that would devastate a young vine barely register. The berries are tiny, thick-skinned, and hyper-concentrated.

■ Years 120–150+ — Some vines are pre-phyloxera — planted before the louse devastated European vineyards in the late 1800s, surviving only because they grow in sandy soils the louse cannot travel through. These vines are irreplaceable. The wines they produce are among the most complex and distinctive on earth. When they die, that wine is gone forever.

Root depth: 15–25m+ (some estimated at 30m+) · Yield: 0.5–1.5 tons/acre · Quality: The pinnacle · Character: Extraordinary concentration, geological complexity, irreplaceable

■ *Famous ancient vines: 150-yr Grenache and Shiraz in Barossa Valley (many ungrafted). 100-yr Zinfandel in Lodi, California. 200-yr País/Carignan in Chile's Maule Valley. Some Douro vines exceed 150 years.*

DECLINE & REPLANTING

Typically 100–150+ Years

■ Eventually, even the greatest vines must be replanted. The decision is rarely simple — phylloxera, disease, economic unviability, or vine death can force it. Replanting means 5–10 years of lost production, then another 25–50 years to return to premium old-vine quality. Some estates replant vine by vine to maintain average age; others rip entire blocks.

■ Climate change is adding new urgency — some estates are replanting with different varieties better adapted to rising temperatures, losing centuries of varietal history in the process. The replanting decision is one of a winemaker's most consequential acts.

The critical question: Is yield still enough to justify hand-harvesting cost? Or is the old vine character worth any price? This decision defines a winery's philosophy.

■ *The greatest tragedy in viticulture: ancient vines lost not to age but to economics. Once a century-old vine is pulled, no amount of money or time can replace what it would have produced.*